

A Topological Model of the Truth-Preservation Incompleteness Theorem: Dissolution of Self-Referential Structure in the Möbius Manifold

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2026-04-30

Abstract

This paper models the requirement of “truth preservation” in formal systems and the associated limitative phenomena (incompleteness) by means of the topological properties of fiber bundles over one-dimensional manifolds. A truth-preservation system that requires strict self-identity ($A = A$) is constructed as a product space (cylinder), while a system that admits isomorphic identity ($A \cong A$) is constructed as a non-orientable space (Möbius strip). We show that the undecidability of self-referential propositions in classical systems can thereby be interpreted geometrically as a global topological obstruction (disconnectedness). Note that this paper does not intend a syntactic generalization of Gödel’s incompleteness theorems; rather, it presents a topological model of limitative phenomena as a concrete example of such a construction.

1 Introduction and Definition of the Geometric Model

We model the propagation of “truth values” in a formal reasoning system as a fiber bundle over the base space S^1 (representing the iterative process of inference). The fiber of the state space is set to $F = \{-1, 1\}$, with 1 corresponding to truth (A) and -1 to falsehood ($\neg A$).

Definition 1.1 (System $T_=$: Strict Truth Preservation). The state space $E_=$ of the system $T_=$, which maintains strict self-identity $A = A$, is defined as the trivial fiber bundle (boundary of a cylinder):

$$E_= = S^1 \times \{-1, 1\}.$$

This space is topologically homeomorphic to two disjoint closed curves $S^1 \sqcup S^1$ and is therefore disconnected ($\pi_0(E_=) = 2$).

Definition 1.2 (System $T_\cong$: Truth Preservation via Isomorphism). The state space $E_\cong$ of the system $T_\cong$, which maintains isomorphic identity $A \cong A$ as a whole, is defined as the non-orientable fiber bundle (boundary of the Möbius strip). That is, it is the quotient space obtained from $[0, 1] \times \{-1, 1\}$ by imposing the equivalence relation $(0, y) \sim (1, -y)$:

$$E_\cong = \frac{[0, 1] \times \{-1, 1\}}{(0, y) \sim (1, -y)}.$$

This space is topologically homeomorphic to a single closed curve S^1 and is therefore connected ($\pi_0(E_\cong) = 1$).

2 Topological Formulation of Limitative Phenomena

We formulate the verification of the self-referential negation proposition G (“this process reaches its own negated state”) within a system as a continuous inference path $\gamma(t)$ along the base space.

Definition 2.1 (Representation of Proposition G via a Continuous Map). We say that proposition G is constructible in a given space if there exists a continuous map $\gamma : [0, 1] \rightarrow E$ starting from the initial state $\gamma(0) = (0, 1)$ and arriving, after a full traversal of the base space, at its own negated state $\gamma(1) = (1, -1)$.

Theorem 2.1 (Boundary of Truth Preservation and Topological Obstruction). (1) In system $T_=$, which requires strict self-identity, the self-referential path G cannot be constructed (is unreachable).
(2) In system $T_\cong$, which requires isomorphic identity, the path G can be constructed, and its structure coincides with a generator of the fundamental group of the space.

Topological Construction of Proof 1. (1) Topological obstruction in system $T_=$ (cylinder):

In the space $E_= = S^1 \times \{-1, 1\}$, the connected component containing initial state $y = 1$ and the component containing state $y = -1$ are completely separated. Therefore, any continuous map satisfying $\gamma(0) = (0, 1)$ must remain within the component $\gamma(t) = (t, 1)$, and no path satisfying $\gamma(1) = (1, -1)$ exists. This shows that the topological boundary imposed by self-identity (disconnectedness) structurally prevents any path from reaching the negated state.

(2) Structural dissolution in system $T_\cong$ (Möbius manifold):

The space $E_\cong$ carries the equivalence relation $(0, 1) \sim (1, -1)$. The path $\gamma(t) = (t, 1)$ proceeds through the local coordinate system while maintaining “truth ($y = 1$)”; however, as $t \rightarrow 1$ after a full traversal of the base space, the global equivalence relation of the space identifies the endpoint geometrically with $(1, -1)$. That is, the continuous local propagation of “truth” is realized globally as “arrival at its own negated state.” This closed curve γ constitutes a generator of the fundamental group $\pi_1(E_\cong) \cong \mathbb{Z}$ of the space $E_\cong$.

3 Conclusion

According to the present model, the self-referential paradoxes and undecidability that constitute limitative phenomena in classical logical systems are interpreted geometrically as topological constraints arising from defining truth preservation as “spatial disconnectedness” (a trivial product space). By extending the condition of truth preservation to continuous isomorphism (a non-orientable manifold), the proposition G that was previously unreachable transforms into a structure describing the global topological property of the space (connectedness with a twist). This model provides an intuitive foundation, from the perspective of topology, for visualizing the relationship between the limits of formal systems and dimensional extension (relaxation of truth-preservation conditions).